# Chapter 11. Resampling Methods

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## 1 Training Error and Test Error

Resampling redraws from the data to learn about an estimator: **cross-validation** and the **bootstrap** estimate prediction error, standard deviation and bias.

**Definition 11.1**

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, $g_{D_n}$ a prediction rule fitted to it, and $L$ a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation $(x_0, y_0)$ independent of $D_n$:

$$PE = \mathrm{E}\big[L(y_0, g_{D_n}(x_0))\big].$$

The training error usually underestimates the test error (the rule is tuned to the same data): a curve through every training point has zero training error but predicts new $y_i^* = f(x_i) + \varepsilon_i^*$ badly. $MS_E = \frac{SS_E}{n-2}$ measures fit, not prediction.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

# 2 Cross-Validation

## 2.1 $K$-fold cross-validation

Split the data randomly into $K$ near-equal parts $C_1, \dots, C_K$ ($n_k$ observations, $n/K$ if divisible); fit on the other $K-1$ parts and predict part $k$:

$$\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation $i$ obtained with part $k$ removed.

## 2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with $(x_i, y_i)$ omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split:

(i) much less bias, since each fit uses $n-1$ observations, almost the whole data set;

(ii) no randomness, since repeated runs give the same result.

**Theorem 11.2: LOOCV shortcut for least squares**

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \left( \frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2,$$

with full-fit $\hat{y}_i$ and leverage $h_{ii}$; $(y_i - \hat{y}_i)/(1 - h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals** (one fit).

**Derivation: proof**

Let $\mathbf{x}_i^\top$ be the $i$th row of $\mathbf{X}$, $A = \mathbf{X}^\top \mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1} \mathbf{x}_i$. Deleting row $i$ gives $A_{(i)} = A - \mathbf{x}_i \mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1} \mathbf{x}_i \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top \mathbf{y}_{(i)} = \mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \left( \mathbf{x}_i^\top A^{-1} + \frac{h_{ii} \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i)}{1 - h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}}$.

**Remark 11.3: Choice of $K$**

LOOCV fits nearly identical training sets, so the errors are highly correlated and their average can be variable; $K = 5$ or 10 is usually better.

### 2.3 Why cross-validation works: model selection

**Example 11.4: Choosing the degree of a polynomial**

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select $p$. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(f(x_i) - \hat{f}_p(x_i))^2,$$

but $f$ is unknown. Consider instead the prediction squared error for new responses $y_i^* = f(x_i) + \varepsilon_i^*$:

$$\begin{aligned} PSE(p) &= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}(y_i^* - \hat{f}_p(x_i))^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*]^2 \\ &= \mathrm{MSE}(p) + \sigma^2, \end{aligned}$$

because $\varepsilon_i^*$ is independent of $\hat{f}_p$ and has mean zero. So minimizing $PSE$ is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{f}_p^{-i}(x_i))^2,$$

where $\hat{f}_p^{-i}$ is fitted without the $i$th point. Since $y_i$ is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}(y_i - \hat{f}_p^{-i}(x_i))^2 = \sigma^2 + \mathrm{E}(f(x_i) - \hat{f}_p^{-i}(x_i))^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$, $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$, so $\hat{p} = \arg\min_p CV(p)$ is sensible; the training error keeps decreasing in $p$.

## 3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

### 3.1 A simple example

Invest $\alpha$ in an asset with return $X$ and $1-\alpha$ in one with return $Y$, minimizing $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

**Theorem 11.5**

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

**Proof.** $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X - Y) \geq 0$, this is a minimum. ■

Plug in sample variances for $\hat{\alpha}$. In a simulation we could draw 1,000 data sets of 100 pairs, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000} \sum_{r=1}^{1000} \hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000 - 1} \sum_{r=1}^{1000} (\hat{\alpha}_r - \bar{\alpha})^2}.$$

### 3.2 Back to the real world

With real data, the bootstrap instead resamples *the original data with replacement* to mimic new samples.

**Definition 11.6: Bootstrap standard error**

1. Draw $Z^{*1}$ of size $n$ from $Z$ with replacement (some observations repeat, some are absent).
2. Compute the estimate $\hat{\alpha}^{*1}$ from $Z^{*1}$.
3. Repeat $B$ times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}$.
4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B - 1} \sum_{r=1}^{B} \left(\hat{\alpha}^{*r} - \bar{\alpha}^*\right)^2}, \qquad \bar{\alpha}^* = \frac{1}{B} \sum_{r'=1}^{B} \hat{\alpha}^{*r'}.$$

**Example 11.7: Three observations**

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\ldots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.

## 4 Selected exercises

**Motivation.** The "63%" of a bootstrap sample (*ISLR*), the limit of the bootstrap SE, LOOCV as prediction-error estimate, and the optimism of the training error.

**Exercise 11.1 (Which observations enter a bootstrap sample)**

A bootstrap sample of size $n$ is drawn with replacement from $n$ observations. (After ISLR, Ch. 5, Exercise 2.)
(a) What is the probability that the first bootstrap observation is not the $j$th original observation? The second?
(b) Show that the probability that the $j$th observation is not in the bootstrap sample is $(1 - 1/n)^n$, and find its limit as $n \to \infty$.
(c) Find the probability that the $j$th observation is in the bootstrap sample for $n = 5, 100, 10{,}000$, and the expected number of distinct original observations in a bootstrap sample.

**Solution.**

(a) Each draw misses observation $j$ with probability $1 - 1/n$; the second draw is independent and from the full set, so again $1 - 1/n$.

(b)

*Step 1.* The $n$ draws are independent, so $P(j \notin Z^*) = \prod_{r=1}^{n} P(\text{draw } r \neq j) = (1 - 1/n)^n$.

*Step 2.* $\log(1-1/n)^n = n\log(1-1/n) = -1 - \frac{1}{2n} - \frac{1}{3n^2} - \cdots \to -1$, using $\log(1-u) = -u - u^2/2 - \cdots$ for $|u| < 1$. Hence $(1 - 1/n)^n \to e^{-1} = 0.368$.

(c)

*Step 1.* $P(j \in Z^*) = 1 - (1 - 1/n)^n$ equals $1 - 0.8^5 = 0.672$ for $n = 5$, $1 - 0.99^{100} = 0.634$ for $n = 100$ and 0.632 for $n = 10{,}000$, close to the limit $1 - e^{-1} = 0.632$.

*Step 2.* The number of distinct observations is $\sum_{j=1}^{n} \mathbf{1}\{j \in Z^*\}$, so by linearity its mean is $n\big[1 - (1 - 1/n)^n\big] \approx 0.632\, n$.

About 37% of the observations are left out of each bootstrap sample; these "out-of-bag" observations can serve as a test set for the rule fitted to that sample. ■

**Exercise 11.2 (Ideal bootstrap standard error of the mean)**

Let $x_1, \ldots, x_n$ be fixed data and $\bar{x}^*$ the mean of a bootstrap sample $x_1^*, \ldots, x_n^*$. Writing $\mathrm{E}^*, \mathrm{Var}^*$ for moments under bootstrap resampling, show that

$$\mathrm{E}^*(\bar{x}^*) = \bar{x}, \qquad \mathrm{Var}^*(\bar{x}^*) = \frac{\hat{\sigma}^2}{n}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2,$$

and conclude that $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ as $B \to \infty$.

**Solution.** [Supplement]

*Step 1.* The $x_r^*$ are iid, each equal to $x_i$ with probability $1/n$. Hence $\mathrm{E}^* x_r^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^* x_r^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$.

*Step 2.* By independence, $\mathrm{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\bar{x}^* = \hat{\sigma}^2/n$.

*Step 3.* The replicates $\bar{x}^{*b}$ are iid from the resampling distribution, so by the LLN $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ a.s.

This plug-in SE $\sqrt{(n-1)/n}\, s/\sqrt{n}$ is slightly below $s/\sqrt{n}$ (divisor $n$); finite-$B$ Monte Carlo error is separate and controllable. ■

**Exercise 11.3 (LOOCV for the sample mean)**

Predict every response by the sample mean, $\hat{f} \equiv \bar{y}$.
(a) Show that $y_i - \bar{y}_{(i)} = \frac{n}{n-1}(y_i - \bar{y})$, where $\bar{y}_{(i)}$ is the mean without observation $i$, and hence $CV_{(n)} = \left(\frac{n}{n-1}\right)^2 \frac{1}{n}\sum_i (y_i - \bar{y})^2$. Check this against the LOOCV shortcut.
(b) If $y_1, \ldots, y_n$ are iid with mean $\mu$ and variance $\sigma^2$, show that $\mathrm{E}\, CV_{(n)} = \sigma^2\left(1 + \frac{1}{n-1}\right)$, the exact prediction error of a mean computed from $n - 1$ observations.

**Solution.** [Supplement]

(a) $\bar{y}_{(i)} = \frac{n\bar{y} - y_i}{n-1}$ gives $y_i - \bar{y}_{(i)} = \frac{n(y_i - \bar{y})}{n-1}$; square and average. With $\mathbf{X} = \mathbf{1}$, $h_{ii} = \frac{1}{n}$, the shortcut gives the same $(y_i - \bar{y})/(1 - \frac{1}{n})$.

(b)

*Step 1.* $\mathrm{E}\sum_i (y_i - \bar{y})^2 = (n-1)\sigma^2$ for iid data (the unbiasedness of $s^2$). Hence $\mathrm{E}\, CV_{(n)} = \frac{n^2}{(n-1)^2} \cdot \frac{(n-1)\sigma^2}{n} = \frac{n\sigma^2}{n-1} = \sigma^2(1 + \frac{1}{n-1})$.

*Step 2.* For a new $y_0$ independent of a sample of size $n-1$ with mean $\bar{y}_{n-1}$, $\mathrm{E}(y_0 - \bar{y}_{n-1})^2 = \mathrm{Var} y_0 + \mathrm{Var}\bar{y}_{n-1} = \sigma^2 + \frac{\sigma^2}{n-1}$, the same value.

So LOOCV is exactly unbiased for the $n-1$-point rule and slightly pessimistic for the $n$-point rule (error $\sigma^2(1 + \frac{1}{n})$). ■

**Exercise 11.4 (Optimism of the training error)**

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{X}$ $n \times p$ of rank $p$, $\mathrm{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}\boldsymbol{\varepsilon} = \sigma^2\mathbf{I}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, and $\mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*$ new responses, $\boldsymbol{\varepsilon}^*$ an independent copy of $\boldsymbol{\varepsilon}$. Show that

$$\mathrm{E}\Big[\frac{1}{n}\|\mathbf{y} - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 - \frac{p}{n}\Big), \qquad \mathrm{E}\Big[\frac{1}{n}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 + \frac{p}{n}\Big).$$

**Solution.** [Supplement]

*Step 1.* (Facts on $\mathbf{H}$.) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\big[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}\mathbf{I}_p = p$. For a random vector $\mathbf{u}$ with mean $\mathbf{0}$ and $\mathrm{Var}\mathbf{u} = \sigma^2\mathbf{I}$ and a fixed matrix $A$, $\mathrm{E}(\mathbf{u}^\top A\mathbf{u}) = \mathrm{tr}\big(A\,\mathrm{E}\mathbf{u}\mathbf{u}^\top\big) = \sigma^2\mathrm{tr}A$.

*Step 2.* (Training error.) $(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, so $\mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ and $\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \boldsymbol{\varepsilon}^\top(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$. By Step 1 its mean is $\sigma^2(n-p)$.

*Step 3.* (Prediction error.) $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\varepsilon}$, so $\mathbf{y}^* - \hat{\mathbf{y}} = \boldsymbol{\varepsilon}^* - \mathbf{H}\boldsymbol{\varepsilon}$ and

$$\mathrm{E}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2 = \mathrm{E}\|\boldsymbol{\varepsilon}^*\|^2 - 2\,\mathrm{E}(\boldsymbol{\varepsilon}^{*\top}\mathbf{H}\boldsymbol{\varepsilon}) + \mathrm{E}(\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}) = n\sigma^2 - 0 + p\sigma^2,$$

the middle term vanishing by independence and $\mathrm{E}\boldsymbol{\varepsilon}^* = \mathbf{0}$.

*Step 4.* Divide by $n$. The gap $2p\sigma^2/n$ grows with $p$: training error falls with model size while prediction error eventually rises, so select models by CV (or a penalty $2p\hat{\sigma}^2/n$). ■